Significance

Craniopharyngiomas (CPs) are rare (incidence of 0.5-2 cases per million people annually), benign yet locally aggressive tumors of the sellar and parasellar regions, with a low histological grade, located near the optic chiasm, hypothalamus, and third ventricle. They primarily affect children aged 5-14 years, with 30%–50% of all cases occurring during childhood and adolescence and are also found in adults aged 50-74 years ([1]).

There are two histologic subtypes: adamantinomatous (ACP) and papillary (PCP). Distinguishing them is clinically important: ACP typically has higher recurrence and worse prognosis, reflecting more infiltrative growth that can compromise vision and hypothalamic–pituitary function, leading to endocrine deficits and cognitive complications ([2]).

The diagnosis of CP is often delayed for years, with the initial symptoms of intracranial pressure. In addition, visual impairment (62%–84%) and endocrine deficits (52%–87%) are common. Endocrine dysfunction results from the disruption of the hypothalamic–pituitary axis, affecting growth hormone (75%), gonadotropins (40%), Adrenocorticotropic Hormones (25%), and Thyroid-Stimulating Hormones (25%); overall, 40%–87% of patients have at least one hormonal deficiency at diagnosis ([3], [4]).

Because of the tendency of the CP to adhere to the nearby optic pathway and surrounding tissues, excision becomes difficult and raises the risk of morbidity and mortality. Some surgeons then prefer partial resection followed by adjuvant post-operative irradiation.

Even though overall survival rates are high (92%), recurrences after complete resection and progressions after incomplete resection can be expected in the range of 71-90%, with radiotherapy reducing it to 21% ([5]).

For surgical planning, various grading systems have been developed based on MRI images and the location of anatomical structures involved with or surrounding the tumors. Due to the anatomical differences among tumors, there is no single preferred approach ([6]). As an example, one recent classification system is the quantitative sensory testing (QST)([7]), which classifies CPs into:

- **Type Q:** CPs that originate from the space below the diaphragm, with an enlarged pituitary fossa
- **Type S:** CPs that originate from the pituitary stalk and tend to spread through cisterns.
- **Type T:** CPs that originate in the top of the pars tuberalis and extend upward to occupy the third ventricle.

Over the past decade, the endonasal endoscopically assisted approach (EEA) has become the standard method for treating most sellar and suprasellar CPs. However, many CPs mainly develop in the infundibulo-tuberal region of the third ventricle (3V). A significant challenge is finding a safe cleavage plane due to the strong attachment between the CP and the hypothalamus, which is especially true for PCPs, as it involves cutting through the functional 3V and poses a risk of damaging the hypothalamus. Other methods, such as through the corpus callosum or opening the lamina terminalis, are used based on preoperative MRI diagnosis of the 3V topography.

AI techniques, such as deep learning models, can provide precise segmentation of CPs and adjacent structures, including the hypothalamus and third ventricle, thereby improving preoperative planning and potentially long-term outcomes ([9]).

References

[1] Walker, D.A., Perilongo, G., Taylor, R.E., & Pollack, I.F. (Eds.). (2020). Brain and Spinal Tumors of Childhood (2nd ed.). CRC Press. https://doi.org/10.1201/9781003090120

[2] Garrè, M. L., & Cama, A. (2007). Craniopharyngioma: modern concepts in pathogenesis and treatment. *Current opinion in pediatrics*, *19*(4), 471–479. https://doi.org/10.1097/MOP.0b013e3282495a22

[3] Daubenbüchel, A. M., & Müller, H. L. (2015). Neuroendocrine Disorders in Pediatric Craniopharyngioma Patients. *Journal of clinical medicine*, *4*(3), 389–413. https://doi.org/10.3390/jcm4030389

[4] Kawamata, T., Kubo, O., & Hori, T. (2005). Histological findings at the boundary of craniopharyngiomas. *Brain tumor pathology*, *22*(2), 75–78. https://doi.org/10.1007/s10014-005-0191-4

[5] Müller H. L. (2011). Diagnostics, treatment, and follow-up in craniopharyngioma. *Frontiers in endocrinology*, *2*, 70. https://doi.org/10.3389/fendo.2011.00070

[6] Lubuulwa, J., & Lei, T. (2016). Pathological and Topographical Classification of 6Craniopharyngiomas: A Literature Review. *Journal of neurological surgery reports*, *77*(3), e121–e127. https://doi.org/10.1055/s-0036-1588060

[7] Bunin, G. R., Surawicz, T. S., Witman, P. A., Preston-Martin, S., Davis, F., & Bruner, J. M. (1998). The descriptive epidemiology of craniopharyngioma. *Journal of neurosurgery*, *89*(4), 547–551. https://doi.org/10.3171/jns.1998.89.4.0547

[8] Pan, J., Qi, S., Liu, Y., Lu, Y., Peng, J., Zhang, X., Xu, Y., Huang, G. L., & Fan, J. (2016). Growth patterns of craniopharyngiomas: clinical analysis of 226 patients. *Journal of neurosurgery. Pediatrics*, *17*(4), 418–433. https://doi.org/10.3171/2015.7.PEDS14449

[9] Pascual, J. M., & Prieto, R. (2022). Craniopharyngioma and the Third Ventricle: This Inescapable Topographical Relationship. *Frontiers in oncology*, *12*, 872689. https://doi.org/10.3389/fonc.2022.872689